

```
In [1]: l = [10, 20, 30, 35, 40, 3, 10, 15, 20, 30, 40, 2.3, 1]
        l
```

```
Out[1]: [10, 20, 30, 35, 40, 3, 10, 15, 20, 30, 40, 2.3, 1]
```

string list slicing

```
In [2]: s1 = 'nit'
        s1
```

```
Out[2]: 'nit'
```

```
In [3]: s1[0]
```

```
Out[3]: 'n'
```

```
In [4]: s1[1]
```

```
Out[4]: 'i'
```

```
In [6]: s1[2]
```

```
Out[6]: 't'
```

```
In [7]: s1[3] # index error
```

```
-----
IndexError                                Traceback (most recent call
l last)
Cell In[7], line 1
----> 1 s1[3]

IndexError: string index out of range
```

```
In [8]: for i in s1:
        print(i)
```

```
n
i
t
```

list slicing

```
In [9]: print(l)
```

```
[10, 20, 30, 35, 40, 3, 10, 15, 20, 30, 40, 2.3, 1]
```

```
In [10]: l[:]
```

```
Out[10]: [10, 20, 30, 35, 40, 3, 10, 15, 20, 30, 40, 2.3, 1]
```

```
In [12]: l[0:8]
```

```
Out[12]: [10, 20, 30, 35, 40, 3, 10, 15]
```

```
In [13]: l[3:]
```

```
Out[13]: [35, 40, 3, 10, 15, 20, 30, 40, 2.3, 1]
```

```
In [14]: l[:7]
```

```
Out[14]: [10, 20, 30, 35, 40, 3, 10]
```

```
In [15]: l[0:20:5]
```

```
Out[15]: [10, 3, 40]
```

```
In [16]: l[3:10:3]
```

```
Out[16]: [35, 10, 30]
```

```
In [17]: l[::-1]
```

```
Out[17]: [1, 2.3, 40, 30, 20, 15, 10, 3, 40, 35, 30, 20, 10]
```

```
In [18]: l[::-2]
```

```
Out[18]: [1, 40, 20, 10, 40, 30, 10]
```

```
In [19]: l[::-3]
```

```
Out[19]: [1, 30, 10, 35, 10]
```

```
In [20]: l[0]
```

```
Out[20]: 10
```

```
In [21]: l[0] = 45
```

```
In [22]: l
```

```
Out[22]: [45, 20, 30, 35, 40, 3, 10, 15, 20, 30, 40, 2.3, 1]
```

```
In [23]: l[-1] = 'nit'  
l
```

```
Out[23]: [45, 20, 30, 35, 40, 3, 10, 15, 20, 30, 40, 2.3, 'nit']
```

```
In [25]: l[-1][0] #nested slicing
```

```
Out[25]: 'n'
```

```
In [26]: print(l[-1][0])
```

n

```
In [27]: 45 in l
```

```
Out[27]: True
```

```
In [28]: 1000 in l
```

```
Out[28]: False
```

```
In [29]: 1 in l
```

```
Out[29]: False
```

```
In [31]: for i in l:  
         print(i)
```

```
45  
20  
30  
35  
40  
3  
10  
15  
20  
30  
40  
2.3  
nit
```

Enumerate

```
In [33]: for i in enumerate(l):  
         print(i)
```

```
(0, 45)  
(1, 20)  
(2, 30)  
(3, 35)  
(4, 40)  
(5, 3)  
(6, 10)  
(7, 15)  
(8, 20)  
(9, 30)  
(10, 40)  
(11, 2.3)  
(12, 'nit')
```

```
In [34]: l1 = [45,30,20,15,10,2.3,'nit']  
         l3 = [100,4,3]
```

```
In [35]: all(l3) # in these list if you dont have 'zero' True will print.
```

```
Out[35]: True
```

```
In [36]: any(l3)
```

```
Out[36]: True
```

```
In [37]: l3.append(0)  
l3
```

```
Out[37]: [100, 4, 3, 0]
```

```
In [38]: all(l3)
```

```
Out[38]: False
```

```
In [39]: any(l3)
```

```
Out[39]: True
```

```
In [ ]:
```